Symmetry Groups of the Plane

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Introduction & motivation

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- Discuss **symmetry groups**:
  - Succinct way to describe an object's symmetry independent of its particulars
  - Identify and classify common symmetries across objects
  - Generate transformations of the plane from those symmetries

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It follows directly from these **group axioms** that the identity and inverses are unique.

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- The integers $\mathbb{Z}$ modulo n, notated $\mathbb{Z}/n\mathbb{Z}$ and consisting of the integers $\{0, 1, \ldots, n-1\}$ is a group under modular addition.

$\mathbb{Z}/n\mathbb{Z}$ provides a useful model for understanding cyclic groups of rotations, as we will find later.

Group homomorphisms

Definition

A map $\varphi : G \to H$ between two groups $(G, \circ)$ and $(H, *)$ is a **group homomorphism** if it respects the binary operation of both groups; that is, for a and b in G, it is always true that

$$\varphi(a \circ b) = \varphi(a) * \varphi(b).$$

If φ is also a bijection, it is called a **group isomorphism** and G and H are said to be **isomorphic**, notated $G \cong H$.

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Groups which are isomorphic are essentially "the same" – that is, they share a common underlying structure.

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- On the plane, symmetries can be broadly classified into reflectional, rotational, translational, and glide reflection symmetries.
- The latter two only occur in "infinite" objects, such as tilings of the plane.

Cyclic groups

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Theorem

If $G = \left\{e, g, g^2, \ldots, g^{n-1}\right\}$ is a finite cyclic group with order n, $G \cong \mathbb{Z}/n\mathbb{Z}$.

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Definition

The cyclic group of all rotations by $360/n$ degrees for a positive integer n is denoted as C_n. Each C_n consists of the members

$$\left\{R_0, R_{360/n}, R_{2\cdot 360/n}, \ldots, R_{(n-1)\cdot 360/n}\right\},$$

where each R_i denotes a rotation by i°.

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For $n \geq 3$, the dihedral groups can be understood as the symmetry groups of regular n-gons.

Examples

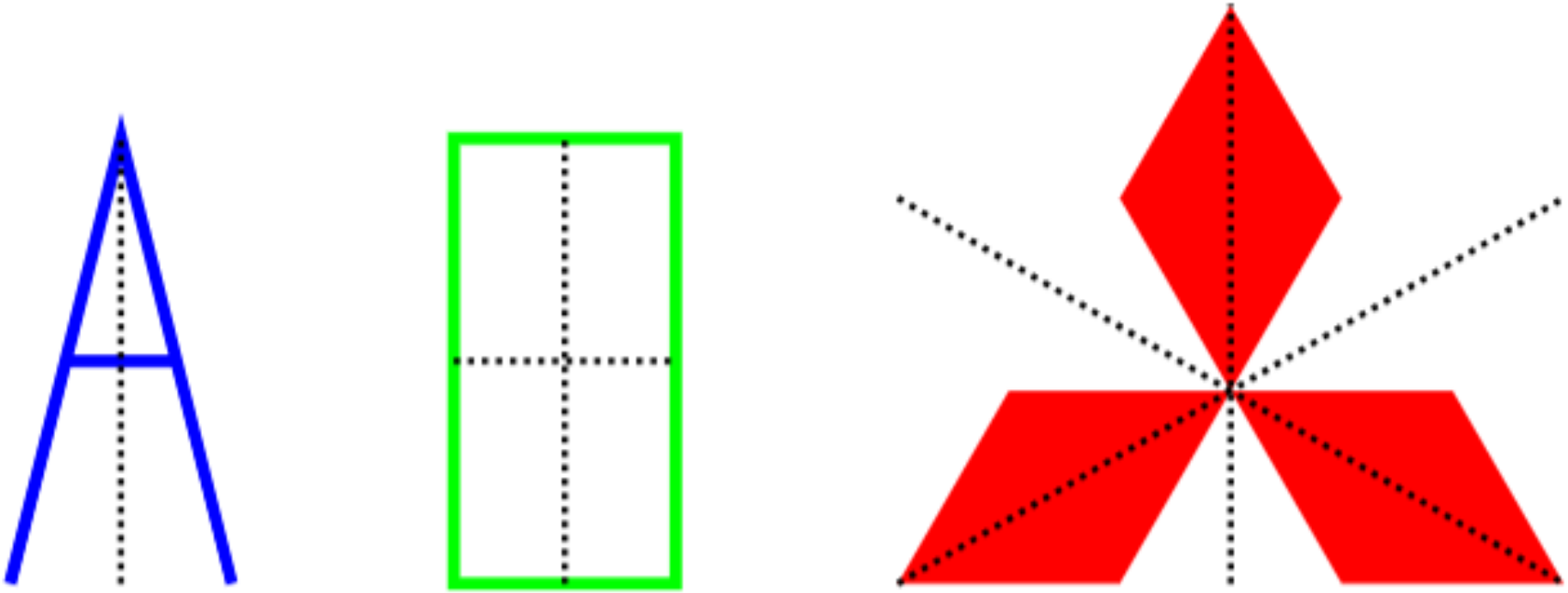

Figure 2: The letter "A" exhibits D_1 symmetry, a non-square rectangle exhibits D_2 symmetry, and the Mitsubishi logo exhibits D_3 symmetry. The dotted lines indicate the axes of symmetry for each figure.

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  - This is a direct consequence of the fact that the composition of two reflections is equivalent to a rotation
- Each D_n has as a subgroup C_n
  - The combination of non-cyclic reflections with a single cyclic subgroup is the defining characteristic of the dihedral groups

Automorphisms and group representations

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The map $\varphi : G \to \mathrm{Aut}(X)$, where X is any set and G is a group, is a **representation of G on X** if it is a group homomorphism.

This means that *a representation of a symmetry group G on $\mathbb{R}^2$ necessarily maps group members modeling transformations to operators performing those transformations*.

Example: D_4 on $\mathbb{R}^2$

- We will construct a representation $\varphi : D_4 \to \mathrm{GL}(\mathbb{R}^2)$ based on the symmetries of the square with vertices at $(-1, 1)$, $(1, 1)$, $(1, -1)$, and $(-1, -1)$.

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- This boils down to finding eight 2×2 matrices under which this square is invariant and assigning each to a member of D_4 in a way that respects the underlying group structure.

Example: D_4 on $\mathbb{R}^2$

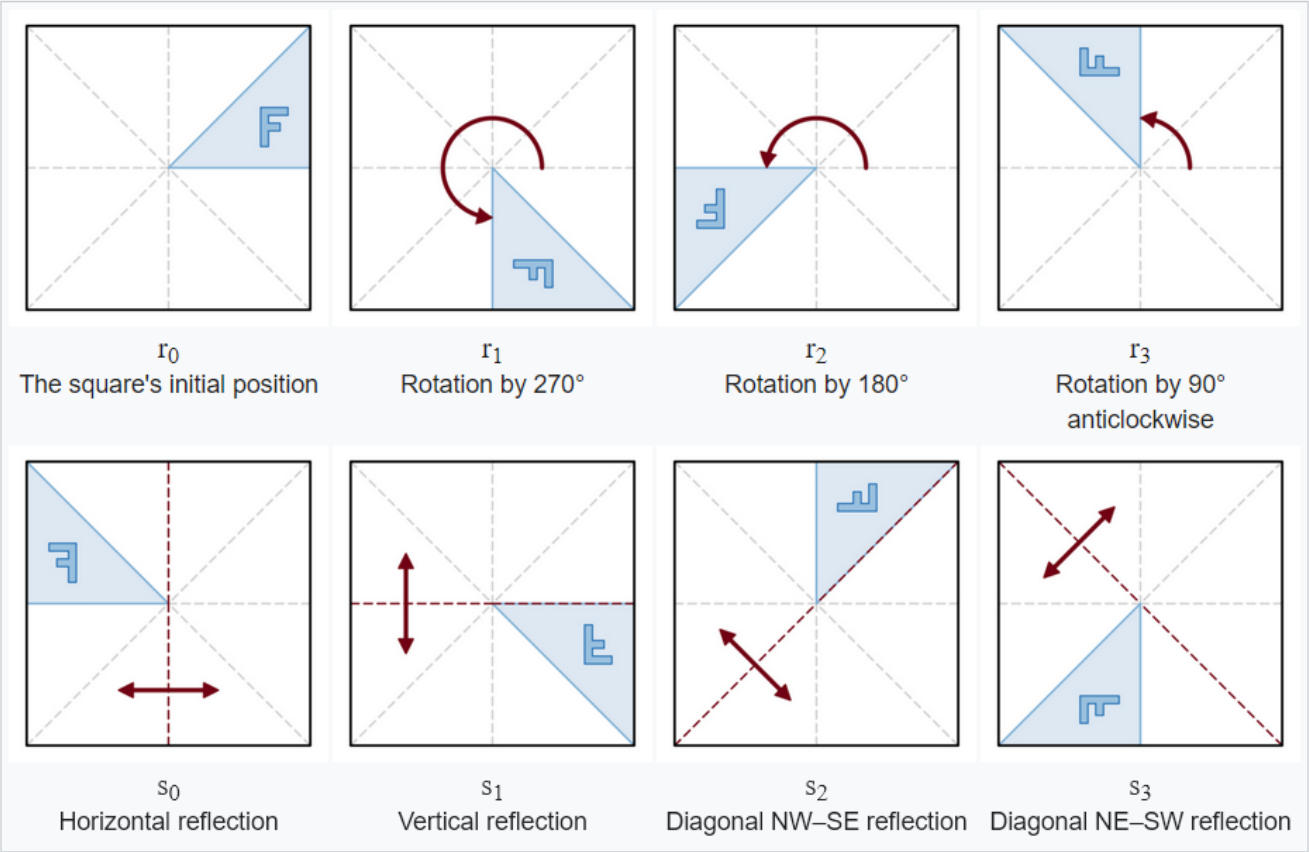
Unsurprisingly, this results in the rotations being mapped to rotation matrices and reflections to reflection matrices:

$$\varphi(R_0) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad \varphi(S_0) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\varphi(R_1) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \qquad \varphi(S_1) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\varphi(R_2) = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \qquad \varphi(S_2) = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\varphi(R_3) = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \qquad \varphi(S_3) = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$



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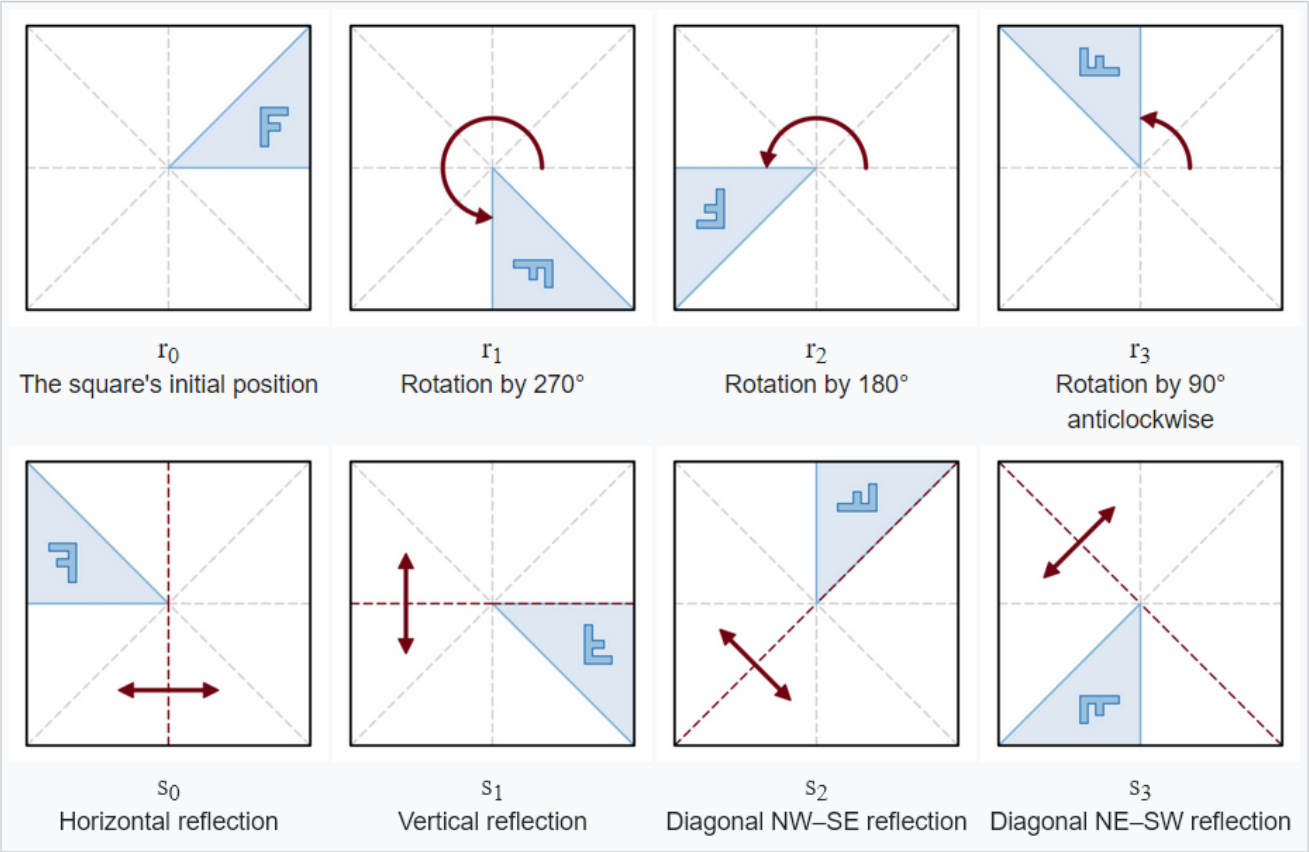

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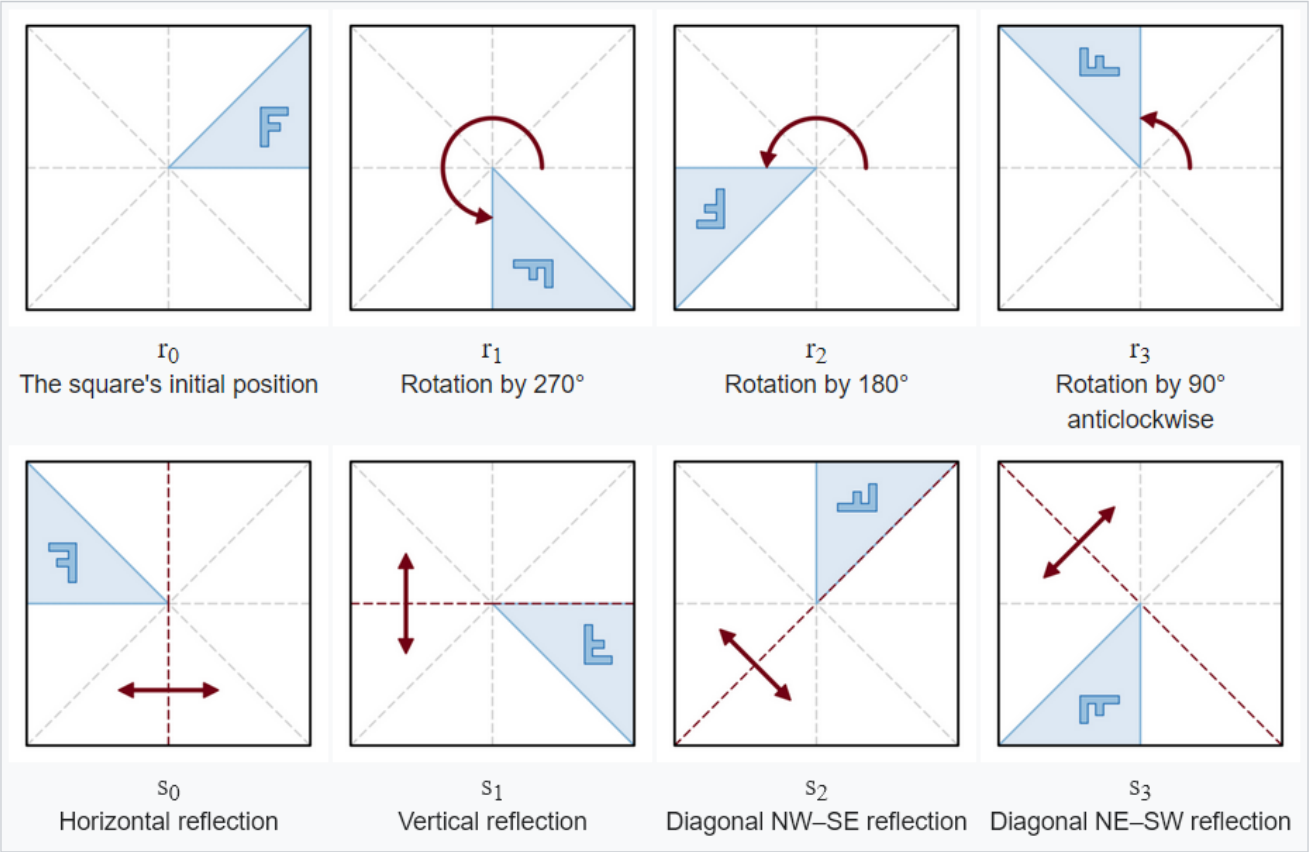

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The flexibility of these representations allows us to conclude that *symmetry is preserved under invertible linear transformations*

Thank you!